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CONTROL DEVICE FOR EXCAVATOR

Abstract

A control device for an excavator is disclosed. The excavator includes an upper slewing body, an attachment provided on the upper slewing body, a slewing mechanism, a lower traveling body, and a blade provided on the lower traveling body. The control device includes circuitry configured to receive an input of an operation command for instructing an operation of the attachment, determine whether a person is detected around the excavator, and cause the excavator to perform an operation of lowering the blade when the person is not detected.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application filed under 35 U.S.C. 111(a) claiming benefit under 35 U.S.C. 120 and 365(c) of PCT International Application No. PCT/JP2023/040560, filed on Nov. 10, 2023, and designating the U.S., which claims priority to Japanese Patent Application No. 2022-187326 filed on Nov. 24, 2022. The entire contents of the foregoing applications are incorporated herein by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a control device for an excavator.

Description of Related Art

[0003] Conventionally, an excavator provided with a blade for performing a ground leveling operation is known. Further, in this excavator, the blade is in contact with the ground when the excavation work is performed, thereby preventing the excavator from falling.

SUMMARY

[0004] According to an embodiment of the present disclosure, a control device for an excavator is provided. The excavator includes an upper slewing body, an attachment provided on the upper slewing body, a slewing mechanism, a lower traveling body, and a blade provided on the lower traveling body. The control device includes: [0005] circuitry configured to receive an input of an operation command for instructing an operation of the attachment, determine whether a person is detected around the excavator, and cause the excavator to perform an operation of lowering the blade when the person is not detected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a side view illustrating a small slewing excavator.

[0007] FIG. 2 is a top view illustrating the small slewing excavator.

[0008] FIG. 3 is a diagram illustrating a configuration example of a hydraulic circuit mounted on the excavator.

[0009] FIG. 4A is a diagram illustrating an extracted portion of a hydraulic system relating to operation of an arm cylinder.

[0010] FIG. 4B is a diagram illustrating an extracted portion of the hydraulic system relating to operation of a boom cylinder.

[0011] FIG. 4C is a diagram illustrating an extracted portion of the hydraulic system relating to operation of a bucket cylinder.

[0012] FIG. 4D is a diagram illustrating an extracted portion of the hydraulic system relating to operation of a Slewing hydraulic motor.

[0013] FIG. 5 is a block diagram illustrating an example of a configuration related to a machine guidance function and a machine control function of the excavator.

[0014] FIG. 6 is a flowchart illustrating operation of the excavator.

[0015] FIG. 7A is a view illustrating a situation when a blade is lowered.

[0016] FIG. 7B is another view illustrating a situation when a blade is lowered.

EMBODIMENT OF THE INVENTION

[0017] In the above-described conventional excavator, an operation of lowering the blade is performed so as to bring the blade into contact with the ground every time the excavation work is performed. Further, in the case of the conventional excavator, when the excavator travels after performing the excavation work, an operation of raising the blade is required in order to protect the traveling surface. Therefore, in the conventional excavator, the operation of lowering and raising the blade is required every time the excavation work is performed, which is troublesome.

[0018] In view of the above-described point, it is an object of the present disclosure to improve operability.

[0019] According to an aspect of embodiments, operability can be improved.

[0020] Embodiments of the present disclosure will be described with reference to the accompanying drawings. FIG. 1 is a side view illustrating a small slewing excavator, and FIG. 2 is a top view illustrating the small slewing excavator.

[0021] Hereinafter, in the present specification, the small slewing excavator may be simply referred to as an “excavator”. An upper slewing body 3 is mounted on a lower traveling body 1 of the excavator 100 via a slewing mechanism 2. A boom 4 is attached to the upper slewing body 3. An arm 5 is attached to the distal end of the boom 4, and a bucket 6 as an end attachment is attached to the distal end of the arm 5. A slope bucket, a dredging bucket, or the like may be used as the end attachment.

[0022] The boom 4, the arm 5, and the bucket 6 constitute an attachment as an example of an attachment, and are hydraulically driven by a boom cylinder 7, an arm cylinder 8, and a bucket cylinder 9, respectively.

[0023] The upper slewing body 3 is provided with a cabin 10 and a power source such as an engine 11. In the cabin 10, an operator's seat, various operation devices 26 required for the excavator 100, and a controller 30 configured to control the driving of the excavator 100 are installed.

[0024] In FIG. 1, when a piston 9a of the bucket cylinder 9 mounted on the arm 5 is installed on a surface at the same height as the installation surface of the crawler, an uppermost portion Th1 of a handrail 70 mounted on a fuel tank 24 is at the highest position in the excavator 100. That is, the uppermost portion Th1 of the handrail 70 is higher than an uppermost portion Th2 of a handrail 60, an uppermost portion Tc of the cabin 10, and a pipe Ta to the arm cylinder 8 mounted on the boom 4. The upper surface of the fuel tank 24 is higher than the muffler cover 90.

[0025] When the handrail 70 is rotated as described later, the uppermost portion Th1 of the handrail 70 is lower than any of the uppermost portion Th2 of the handrail 60, the uppermost portion Tc of the cabin 10, and the pipe Ta to the arm cylinder 8 mounted on the boom 4.

[0026] In the excavator 100 according to the present embodiment, the lower traveling body 1 is provided with a blade 95 used for ground leveling work or the like. The blade 95 has a tip portion 95a and a support portion 95b, and the blade is driven by a blade cylinder (not illustrated) provided in the lower traveling body 1. More specifically, the blade cylinder extends and contracts in response to the operation of the operator to move the blade 95 up and down. In other words, the excavator 100 performs a blade lowering operation of lowering the blade 95 and a blade raising operation of raising the blade 95 in response to the operation of the operator.

[0027] The excavator 100 also includes a blade angle sensor 96 for detecting the angle of the blade 95. The angle of the blade 95 may be, for example, an angle with respect to the horizontal plane of the support portion 95b.

[0028] The blade angle sensor 96 of the present embodiment transmits a detected signal to the controller 30 of the excavator 100 by wireless communication. The blade angle sensor 96 of the present embodiment may transmit a signal by electric power supplied by vibration power generation, solar power generation, or the like.

[0029] When the excavator 100 according to the present embodiment receives an input of an operation for instructing an operation of the attachment, the controller 30 performs a blade

lowering operation for lowering the blade **95**, and causes the blade **95** to contact the ground, thereby prohibiting the excavator **100** from traveling.

[0030] When the operation for the attachment is ended and the operation for instructing the traveling is received, the excavator **100** according to the present embodiment performs the blade raising operation for raising the blade **95** by the controller **30**. In other words, the controller **30** performs the blade raising operation when no operation command is input to the attachment and a travel command for instructing the excavator to travel is input.

[0031] In the present embodiment, since the blade **95** is controlled by the controller **30** in this manner, the operator does not need to perform an operation of lowering the blade **95** every time the excavation work is performed, and the operation is thus simplified.

[0032] In the present embodiment, when the excavation work is performed, the blade **95** is lowered and is brought into contact with the ground, and thus it is possible to improve the stability of the body of the excavator **100** during the excavation work. Further, in the present embodiment, in a state where the blade **95** is in contact with the ground, the controller **30** prohibits the excavator **100** from traveling. Therefore, it is possible to prevent the excavator **100** from traveling with the blade **95** being lowered. The processing of the controller **30** will be described in detail later.

[0033] In the excavator **100** according to the present embodiment, as illustrated in FIG. 2, the boom **4** is rotatably supported near the center of the upper slewing body **3**. The cabin **10** is installed on the front side of the upper slewing body **3** and on the left side of the boom **4**. A urea water tank cover **19** for covering a urea water tank (described later) is attached to the front side of the upper slewing body **3** and the right side of the boom **4**. The fuel tank **24** and a hydraulic fluid tank **27** are disposed behind the urea water tank cover **19**. The fuel tank **24** is disposed on the outer side, and the hydraulic fluid tank **27** is disposed on the inner side of the fuel tank **24**.

[0034] The upper surface of the urea water tank cover **19** and the upper surface of the hydraulic fluid tank **27** are used as a passage for a worker to pass during maintenance or the like. On the other hand, the upper surface of the fuel tank **24** is located at a position higher than the upper surface of the hydraulic fluid tank **27** as described later, and is not used as a passage for the operator to pass.

[0035] A handrail **60** is attached to the outer peripheral portion of the upper slewing body **3** from the urea water tank cover **19** to the fuel tank **24**. The handrail **60** is a rail for protecting the worker from falling when the worker climbs on the upper slewing body **3**. The upper end of the handrail **60** extends to the end of the fuel tank **24**. A handrail **70** extends from the position where the handrail **60** terminates in the direction in which the handrail **60** extends. The handrail **70** is attached to the upper surface of the fuel tank **24** and is bent inward along the corner of the upper surface of the fuel tank **24**. The handrail **70** is provided at a position where the operator can hold the handrail **70** when the operator climbs the upper surface of the hydraulic fluid tank **27**.

[0036] A muffler cover **90** for covering an exhaust gas treatment device described later is provided behind the fuel tank **24** and the hydraulic fluid tank **27**. An exhaust pipe (muffler) **92** extending from the exhaust gas treatment device protrudes from a portion of the muffler cover **90** near the fuel tank **24**.

[0037] The urea water tank cover **19**, the fuel tank **24**, the hydraulic fluid tank **27**, the handrail **60**, the handrail **70**, and the muffler cover **90** are disposed on the right side of the upper slewing body **3**.

[0038] As described above, the boom **4** is rotatably attached to the center of the upper slewing body **3**. The shape of the rear portion of the upper slewing body **3** is defined by the slewing radius of the excavator, and is an arc shape having the slewing radius. A counterweight **28** is disposed at the center of the rear portion of the upper slewing body **3**. The engine **11** is disposed and fixed in a space between the counterweight **28** and the boom **4**. In FIGS. 1 and 2, the engine **11** is not illustrated because an engine hood **35** is provided to cover the upper portion of the engine **11**.

[0039] A passage **32** is provided on the front side of the engine hood **35**, and an operator passes the

passage **32** when the engine hood **35** is opened to perform maintenance work of the engine **11**. A working scaffold **34** is provided at a position lower than the passage **32** on the front side of the passage **32**. The working scaffold **34** is provided so as to cover the slewing motor disposed beside the hydraulic fluid tank **27**, and also functions as a slewing motor cover. When the worker climbs the upper passage **32**, the worker first climbs the lower working scaffold **34** and then climbs the higher passage **32**. At this time, the worker can easily climb from the working scaffold **34** to the passage **32** by placing his/her hand on a portion of the end portion of the handrail **70** extending in the vertical direction.

[0040] The excavator **100** according to the present embodiment further includes a boom angle sensor **S1**, an arm angle sensor **S2**, a bucket angle sensor **S3**, a body inclination sensor **S4**, and a slewing state sensor **S5**. The excavator **100** according to the present embodiment includes a space recognition device **81**, an orientation detection device **82**, an input device **83**, a positioning device **84**, a display device **D1**, and a sound output device **D2**.

[0041] The boom angle sensor **S1** is attached to the boom **4** and detects an elevation angle of the boom **4** with respect to the upper slewing body **3** (hereinafter, referred to as a “boom angle”), for example, an angle formed by a straight line connecting fulcrums at both ends of the boom **4** with respect to a slewing plane of the upper slewing body **3** in a side view.

[0042] The boom angle sensor **S1** may include, for example, a rotary encoder, an accelerometer, a gyro sensor (angular velocity sensor), a six axis sensor, an inertial measurement unit (IMU), and the like. Hereinafter, the same applies to the arm angle sensor **S2**, the bucket angle sensor **S3**, and the body inclination sensor **S4**. A detection signal corresponding to the boom angle by the boom angle sensor **S1** is incorporated into the controller **30**.

[0043] The arm angle sensor **S2** is attached to the arm **5** and detects a rotation angle of the arm **5** with respect to the boom **4** (hereinafter, referred to as an “arm angle”), for example, an angle formed by a straight line connecting fulcrums at both ends of the arm **5** with respect to a straight line connecting fulcrums at both ends of the boom **4** in a side view. A detection signal corresponding to the arm angle by the arm angle sensor **S2** is incorporated into the controller **30**.

[0044] The bucket angle sensor **S3** is attached to the bucket **6** and detects a rotation angle of the bucket **6** with respect to the arm **5** (hereinafter, referred to as a “bucket angle”), for example, an angle formed by a straight line connecting a fulcrum and a tip (a blade edge) of the bucket **6** with respect to a straight line connecting fulcrums at both ends of the arm **5** in a side view. A detection signal corresponding to the bucket angle by the bucket angle sensor **S3** is incorporated into the controller **30**.

[0045] In the present embodiment, the above-described various angle sensors may include an operation amount detection part that detects an operation amount, and the various angle sensors may calculate an angle based on the detected operation amount.

[0046] The body inclination sensor **S4** detects an inclination state of the body (e.g., the upper slewing body **3**) with respect to the horizontal plane. The body inclination sensor **S4** is attached to, for example, the upper slewing body **3**, and detects inclination angles (hereinafter, referred to as a “front-rear inclination angle” and a “left-right inclination angle”) around two axes in the front-rear direction and the left-right direction of the excavator **100** (i.e., the upper slewing body **3**). The body inclination sensor **S4** may include, for example, an accelerometer, a gyroscope (angular velocity sensor), a six axis sensor, an IMU, and the like. The controller **30** receives a detection signal corresponding to the inclination angle (the front-rear inclination angle and the left-right inclination angle) from the body inclination sensor **S4**.

[0047] The slewing state sensor **S5** is attached to the upper slewing body **3** and outputs detection information related to the slewing state of the upper slewing body **3**. The slewing state sensor **S5** detects, for example, a slewing angular velocity and a slewing angle of the upper slewing body **3**. The slewing state sensor **S5** includes, for example, a gyro sensor, a resolver, a rotary encoder, and the Like. Note that, when the body inclination sensor **S4** includes a gyro sensor, a six axis sensor,

an IMU, or the like capable of detecting angular velocities about three axes, the slewing state (e.g., slewing angular velocities) of the upper slewing body **3** may be detected based on the detection signal of the body inclination sensor **S4**. In this case, the slewing state sensor **S5** may be omitted. [0048] The space recognition device **81** is configured to recognize an object existing in a three dimensional space around the excavator **100** and measure (calculate) a positional relationship such as a distance from the space recognition device **81** or the excavator **100** to the recognized object. The space recognition device **81** may include, for example, an ultrasonic sensor, a millimeter wave radar, a monocular camera, a stereo camera, a light detecting and ranging (LIDAR), a range image sensor, and an infrared sensor.

[0049] In the present embodiment, the space recognition device **81** includes a front recognition sensor **81F** attached to the front end of the upper surface of the cabin **10** and a rear recognition sensor **81B** attached to the rear end of the upper surface of the upper slewing body **3**. The space recognition device **81** of the present embodiment may include a left recognition sensor attached to the left end of the upper surface of the upper slewing body **3** and a right recognition sensor attached to the right end of the upper surface of the upper slewing body **3**. Further, an upper recognition sensor that recognizes an object existing in the space above the upper slewing body **3** may be attached to the excavator **100**.

[0050] The orientation detection device **82** detects information (e.g., a slewing angle of the upper slewing body **3** with respect to the lower traveling body **1**) related to a relative relationship between the orientation of the upper slewing body **3** and the orientation of the lower traveling body **1**.

[0051] The orientation detection device **82** may include, for example, a combination of a geomagnetic sensor attached to the lower traveling body **1** and a geomagnetic sensor attached to the upper slewing body **3**. The orientation detection device **82** may include a combination of a GNSS receiver attached to the lower traveling body **1** and a GNSS receiver attached to the upper slewing body **3**. The orientation detection device **82** may include a rotary encoder, a rotary position sensor, or the like capable of detecting the relative slewing angle of the upper slewing body **3** with respect to the lower traveling body **1**, that is, the slewing state sensor **S5** described above, and may be attached to, for example, a center joint provided in association with the slewing mechanism **2** that implements the relative rotation between the lower traveling body **1** and the upper slewing body **3**.

[0052] The orientation detection device **82** may include a camera attached to the upper slewing body **3**. In this case, the orientation detection device **82** performs known image processing on an image (input image) captured by the camera attached to the upper slewing body **3**, thereby detecting an image of the lower traveling body **1** included in the input image.

[0053] In the case of a configuration in which the upper slewing body **3** is slewably driven by a motor instead of the slewing hydraulic motor **2A**, the orientation detection device **82** may be a resolver.

[0054] The input device **83** is provided within reach of an operator seated in the cabin **10**, receives various operation inputs from the operator, and outputs signals corresponding to the operation inputs to the controller **30**. For example, the input device **83** may include a touch panel mounted on a display of the display device **D1** that displays various information images.

[0055] Further, for example, the input device **83** can include a button switch, a lever, a toggle, or the like installed around the display device **D1**. The input device **83** may include a knob switch provided on the operation device **26** (e.g., a switch **SW** provided on a left operation lever **26L**). A signal corresponding to the operation content on the input device **83** is incorporated into the controller **30**.

[0056] The switch **SW** is, for example, a push button switch provided at the tip of the left operation lever **26L**. The operator can operate the left operation lever **26L** while pressing the switch **SW**. The switch **SW** may be provided on a right operation lever **26R** or at another position in the cabin **10**.

[0057] A positioning device **84** measures the position and orientation of the upper slewing body **3**.

The positioning device **84** is, for example, a global navigation satellite system (GNSS) compass, and detects the position and orientation of the upper slewing body **3**. Detection signals corresponding to the position and orientation of the upper slewing body **3** are incorporated into the controller **30**. Further, the function of detecting the orientation of the upper slewing body **3** among the functions of the positioning device **84** may be replaced by an orientation sensor attached to the upper slewing body **3**.

[0058] The display device **D1** is provided at a position easily visible from the operator seated in the cabin **10**, and displays various information images under the control of the controller **30**. The display device **D1** may be connected to the controller **30** via an in-vehicle communication network such as a controller area network (CAN) or may be connected to the controller **30** via a one-to-one dedicated line.

[0059] The sound output device **D2** is provided in the cabin **10**, for example, is connected to the controller **30**, and outputs sound under the control of the controller **30**. The sound output device **D2** is, for example, a speaker, a buzzer, or the like. The sound output device **D2** outputs various kinds of information in response to an audio command from the controller **30**.

[0060] Next, a configuration example of a hydraulic circuit mounted on the excavator **100** of FIG. **1** will be described with reference to FIG. **3**. FIG. **3** is a diagram illustrating a configuration example of a hydraulic circuit mounted on the excavator. In FIG. **3**, the high-pressure hydraulic line, the pilot line, and the electric control system are indicated by a solid line, a broken line, and a dash dot line, respectively.

[0061] The main pumps **14L** and **14R** are variable displacement hydraulic pumps driven by the engine **11**. In the present embodiment, the main pump **14L** circulates the hydraulic fluid to the hydraulic fluid tank **27** through a center bypass oil passage **21L** passing through each of the control valves **171L** to **175L** constituting the control valve **17**. The main pump **14L** can supply the hydraulic fluid to each of the control valves **172L** to **175L** through a parallel oil passage **22L** extending in parallel to the center bypass oil passage **21L**.

[0062] Similarly, the main pump **14R** circulates the hydraulic fluid to the hydraulic fluid tank **27** through a center bypass oil passage **21R** passing through each of the control valves **171R** to **175R** constituting the control valve **17**. The main pump **14R** can supply the hydraulic fluid to each of the control valves **172R** to **175R** through a parallel oil passage **22R** extending in parallel to the center bypass oil passage **21R**. Hereinafter, the main pump **14L** and the main pump **14R** may be collectively referred to as a “main pump **14**”. The same applies to the other components configured as a pair of left and right components.

[0063] The control valve **171L** is a spool valve that switches the flow of the hydraulic fluid discharged from the main pump **14L** to supply the hydraulic fluid to the left traveling hydraulic motor **1A** when a left travel lever (not illustrated) is operated.

[0064] The control valve **171R** is a spool as a straight traveling control valve. In the present embodiment, the control valve **171R** has a first position and a second position. Specifically, the first position includes a flow passage that communicates the main pump **14L** with the parallel oil passage **22L** and a flow passage that communicates the main pump **14R** with the control valve **172R**. The second position includes a flow passage that communicates the main pump **14R** with the parallel oil passage **22L** and a flow passage that communicates the main pump **14L** with the control valve **172R**.

[0065] The control valve **172L** is a spool value that switches the flow of the hydraulic fluid to supply the hydraulic fluid discharged from the main pump **14L** to a blade cylinder **95A** when the blade **95** is operated by the operation device **26**. In other words, the control valve **172L** is a spare control valve.

[0066] The control valve **172R** is a spool value that switches the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14** to the right traveling hydraulic motor **1B** when the operation device **26** is operated. The operation device **26** will be described in detail

later.

[0067] The control valve **173L** is a spool valve that switches the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14** to the slewing hydraulic motor **2A** when the slewing mechanism **2** is operated by the operation device **26**.

[0068] The control valve **173R** is a spool valve that switches the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14R** to the bucket cylinder **9** when the bucket **6** is operated by the operation device **26**.

[0069] The control valves **174L** and **174R** are spool valves that switch the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14** to the boom cylinder **7** when the boom **4** is operated by the operation device **26**. The control valve **174L** additionally supplies the hydraulic fluid to the boom cylinder **7** when the operation lever for operating the boom **4** is operated in the boom raising direction by a predetermined lever operation amount or more. The operation lever is part of the operation device **26**.

[0070] The control valves **175L** and **175R** are spool valves that switch the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14** to the arm cylinder **8** when the arm **5** is operated by the operation device **26**. The control valve **175R** additionally supplies the hydraulic fluid to the arm cylinder **8** when the operation lever for operating the arm **5** is operated by a predetermined lever operation amount or more.

[0071] The hydraulic fluid flowing out from each of the left traveling hydraulic motor **1A**, the blade cylinder **95A**, the slewing hydraulic motor **2A**, and the arm cylinder **8** is discharged to the hydraulic fluid tank **27** through the return oil passage **23L**. Similarly, the hydraulic fluid flowing out from each of the right traveling hydraulic motor **1B**, the bucket cylinder **9**, and the boom cylinder **7** is discharged to the hydraulic fluid tank **27** through the return oil passage **23R**. Further, a part of the hydraulic fluid flowing out from the arm cylinder **8** may be discharged to the hydraulic fluid tank **27** through a return oil passage **23R**.

[0072] The center bypass oil passages **21L** and **21R** are provided with negative control throttles **20L** and **20R** between the control valves **175L** and **175R** on the most downstream side and the hydraulic fluid tank **27**, respectively. Hereinafter, the negative control is abbreviated as “negative control”. The negative control throttles **20L** and **20R** restrict the flow of the hydraulic fluid discharged from the main pumps **14L** and **14R** to generate a negative control pressure upstream of the negative control throttles **20L** and **20R**.

[0073] The controller **30** performs the negative control by using the negative control pressure. Specifically, the lower the negative control pressures generated in the negative control throttles **20L** and **20R**, the more the discharge amounts of the main pumps **14L** and **14R** is increased. When the negative control pressures generated by the negative control throttles **20L** and **20R** exceed predetermined pressures, the discharge amounts of the main pumps **14L** and **14R** are reduced to predetermined lower limit values.

[0074] The relief valve **50** is a valve for controlling the pressure in the rod side oil chamber of the blade cylinder **95A** to be equal to or lower than predetermined closing relief pressures.

[0075] The load check valve **51** is a valve for preventing the hydraulic fluid in the blade cylinder **95A** from flowing back to the parallel oil passage **22L**.

[0076] The negative control pressures generated upstream of the negative control throttles **20L** and **20R** are detected by the pressure sensors **61L** and **61R**, and the detected values are output to the controller **30** as electrical negative control pressure signals.

[0077] The pressures sensors **62L** and **62R** detect the discharge pressures of the main pumps **14L** and **14R**, and output the detected values to the controller **30** as electric discharge pressure signals.

[0078] The pressure sensor **63** detects the oil pressure in the rod side oil chamber of the blade cylinder **95A** and outputs the detected value to the controller **30** as an electric blade rod pressure signal.

[0079] The pressure sensor **64** is one of the pressure sensors, which detects a pilot hydraulic

pressure applied to the right pilot port of the control valve **172L**, and outputs the detected value to the controller **30** as an electric signal.

[0080] The pressure sensor **65** is one of the pressure sensors, which detects the pilot pressure applied to the left side (boom raising side) pilot port of the control valve **174L** and the right side (boom raising side) pilot port of control valve **174R** (hereinafter referred to as “boom raising pilot pressure”), and outputs the detected value as an electrical boom raising pilot pressure signal to the controller **30**.

[0081] The controller **30** receives outputs from the pressure sensors **61L**, **61R**, **62L**, **62R**, **63**, **64**, **65**, and the like, and causes the CPU to execute a program for adjusting the discharge amounts of the main pumps **14L** and **14R**.

[0082] Further, when both the hydraulic actuators (e.g., the blade cylinder **95A**) related to the main pump **14L** and the hydraulic actuators (e.g., the boom cylinder **7**) related to the main pump **14R** are continuously operated by full lever/full pedal (e.g., an operation amount of 80% or more when a neutral state of the lever/pedal is 0% and a maximum operation state is 100%), the controller **30** sets the discharge amount **L1** of the main pump **14L** and the discharge amount **L2** of the main pump **14R** to be the same. Hereinafter, this method is referred to as a “discharge amount synchronization method”.

[0083] Next, a configuration for the controller **30** to operate the actuators by the machine control function will be described with reference to FIGS. **4A** to **4D**. FIGS. **4A** to **4D** are diagrams in which a part of the hydraulic system is extracted. Specifically, FIG. **4A** is a diagram illustrating an extracted portion of a hydraulic system relating to the operation of the arm cylinder **8**, and FIG. **4B** is a diagram illustrating an extracted portion of the hydraulic system relating to the operation of the boom cylinder **7**. FIG. **4C** is a diagram illustrating an extracted portion of the hydraulic system related to the operation of the bucket cylinder **9** is extracted, and FIG. **4D** is a diagram illustrating an extracted portion of the hydraulic system related to the operation of the slewing hydraulic motor **2A** is extracted.

[0084] As illustrated in FIGS. **4A** to **4D**, the hydraulic system includes proportional valves **31**. The proportional valves **31** include proportional valves **31AL** to **31DL** and proportional valves **31AR** to **31DR**.

[0085] The proportional valves **31** each functions as a control valve for machine control. The proportional valves **31** are disposed in a conduit connecting pilot pump **15** and pilot ports of corresponding control valves in a control valve **17**, and is configured to be able to change a flow passage area of the conduit.

[0086] In the present embodiment, the proportional valves **31** operate in response to a control command output from the controller **30**. Therefore, the controller **30** can supply the hydraulic fluid discharged by the pilot pump **15** to the pilot ports of the corresponding control valves in the control valve **17** via the proportional valves **31**, independent of the operation of the operation device **26** by the operator. The controller **30** can apply the pilot pressures generated by the proportional valves **31** to the pilot ports of the corresponding control valves.

[0087] With this configuration, even when an operation is not performed on a specific operation device **26**, the controller **30** can operate a hydraulic actuator corresponding to the specific operation device **26**. Further, even when an operation is being performed on the specific operation device **26**, the controller **30** can forcibly stop the operation of the hydraulic actuator corresponding to the specific operation device **26**.

[0088] For example, as illustrated in FIG. **4A**, the left operation lever **26L** is used to operate the arm **5**. Specifically, the left operation lever **26L** uses the hydraulic fluid discharged from the pilot pump **15** to apply pilot pressures corresponding to the operation in the front-rear direction to the pilot ports of the control valve **175L** and the control valve **175R**. Specifically, when the left operation lever **26L** is operated in the arm closing direction (rearward direction), the left operation lever **26L** applies a pilot pressure corresponding to the operation amount to the right pilot port of

the control valve **175L** and the left pilot port of the control valve **175R**. When the left operation lever **26L** is operated in the arm opening direction (forward direction), the left operation lever **26L** applies a pilot pressure corresponding to the operation amount to the left pilot port of the control valve **175L** and the right pilot port of the control valve **175R**.

[0089] The operation device **26** is provided with a switch **SW**. In the present embodiment, the switch **SW** includes a switch **SW1** and another switch provided at the tip of a travel lever (not illustrated).

[0090] The switch **SW1** is a push button switch provided at the tip of the left operation lever **26L**. The operator can operate the left operation lever **26L** while pressing the switch **SW1**. The switch **SW1** may be provided on the right operation lever **26R** or may be provided at another position in the cabin **10**.

[0091] The other switch is a push button switch provided at the tip of the left traveling lever. The operator can operate the left travel lever while pressing another switch. The other switch may be provided on the right travel lever included in the operation device **26**, or may be provided at another position in the cabin **10**.

[0092] The operation sensors **29** include operation sensors **29LA** and **29LB**, and operation sensors **29RA** and **29RB**. The operation sensor **29LA** detects the operation content of the left operation lever **26L** in the front-rear direction by the operator, and outputs the detected value to the controller **30**.

[0093] The proportional valve **31AL** operates in response to a control command (electric current command) output from the controller **30**. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the right pilot port of the control valve **175L** and the left pilot port of the control valve **175R** via the proportional valve **31AL**.

[0094] The proportional valve **31AR** operates in response to a control command (electric current command) output from the controller **30**. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the left pilot port of the control valve **175L** and the right pilot port of the control valve **175R** via the proportional valve **31AR**. The proportional valve **31AL** can adjust the pilot pressure so that the control valve **175L** and the control valve **175R** can be stopped at any positions. Similarly, the proportional valve **31AR** can adjust the pilot pressure so that the control valve **175L** and the control valve **175R** can be stopped at any positions.

[0095] With this configuration, the controller **30** can supply the hydraulic fluid discharged by the pilot pump **15** to the right pilot port of the control valve **175L** and the left pilot port of the control valve **175R** via the proportional valve **31AL** in response to the arm closing operation by the operator. Further, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the right pilot port of the control valve **175L** and the left pilot port of the control valve **175R** via the proportional valve **31AL**, independently of the arm closing operation by the operator. That is, the controller **30** can close the arm **5** in response to the arm closing operation by the operator or independently of the arm closing operation by the operator.

[0096] Further, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the left pilot port of the control valve **175L** and the right pilot port of the control valve **175R** via the proportional valve **31AR** in response to the arm opening operation by the operator. Further, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the left pilot port of the control valve **175L** and the right pilot port of the control valve **175R** via the proportional valve **31AR**, independently of the arm opening operation by the operator. That is, the controller **30** can open the arm **5** in response to the arm opening operation by the operator or independently of the arm opening operation by the operator.

[0097] Further, with this configuration, even when the arm closing operation is performed by the operator, the controller **30** can reduce the pilot pressures applied to the pilot ports on the closing side of the control valves **175** (the left pilot port of the control valve **175L** and the right pilot port of the control valve **175R**) as necessary, and forcibly stop the closing operation of the arm **5**. The

same applies to a case where the opening operation of the arm 5 is forcibly stopped when the arm opening operation is performed by the operator.

[0098] Alternatively, even when the operator is performing the arm closing operation, the controller 30 may control the proportional valve 31AR as necessary to increase the pilot pressures applied to the pilot ports on the opening side of the control valves 175 (the right pilot port of the control valve 175L and the left pilot port of the control valve 175R) opposite to the pilot ports on the closing side of the control valves 175, and forcibly return the control valves 175 to the neutral positions, thereby forcibly stopping the closing operation of the arm 5. The same applies to a case where the opening operation of the arm 5 is forcibly stopped when the arm opening operation is performed by the operator.

[0099] Although description will be omitted with reference to FIGS. 4B to 4D below, the same applies to a case where the operation of the boom 4 is forcibly stopped when the boom raising operation or the boom lowering operation is performed by the operator, a case where the operation of the bucket 6 is forcibly stopped when the bucket closing operation or the bucket opening operation is performed by the operator, and a case where the slewing operation of the upper slewing body 3 is forcibly stopped when the slewing operation is performed by the operator. The same also applies to a case where the traveling operation of the lower traveling body 1 is forcibly stopped when the traveling operation is performed by the operator.

[0100] As illustrated in FIG. 4B, the right operation lever 26R is used to operate the boom 4. Specifically, the right operation lever 26R uses the hydraulic fluid discharged from the pilot pump 15 to apply pilot pressures corresponding to the operation in the front-rear direction to the pilot port of the control valve 174L and the pilot port of the control valve 174R. Specifically, when the right operation lever 26R is operated in the boom raising direction (rearward direction), the right operation lever 26R applies a pilot pressure corresponding to the operation amount to the right pilot port of the control valve 174L and the left pilot port of the control valve 174R. When the right operation lever 26R is operated in the boom lowering direction (forward direction), the right operation lever 26R applies pilot pressure corresponding to the operation amount to the right pilot port of the control valve 174R.

[0101] The operation sensor 29RA detects the operation content of the right operation lever 26R in the front-rear direction by the operator, and outputs the detected value to the controller 30.

[0102] The proportional valve 31BL operates in response to a control command (electric current command) output from the controller 30. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump 15 to the right pilot port of the control valve 174L and the left pilot port of the control valve 174R via the proportional valve 31BL. The proportional valves 31BR operate in response to a control command (electric current command) output from the controller 30.

[0103] The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump 15 to the right pilot port of the control valve 174R via the proportional valve 31BR. The proportional valve 31BL can adjust the pilot pressure so that the control valve 174L and the control valve 174R can be stopped at any positions. The proportional valve 31BR can adjust the pilot pressure so that the control valve 174R can be stopped at any positions.

[0104] With this configuration, the controller 30 can supply the hydraulic fluid discharged by the pilot pump 15 to the right pilot port of the control valve 31BL and the left pilot port of the control valve 174L via the proportional valves 174R in response to the boom raising operation by the operator. The controller 30 can supply the hydraulic fluid discharged from the pilot pump 15 to the right pilot port of the control valve 174L and the left pilot port of the control valve 174R via the proportional valve 31BL, independently of the boom raising operation by the operator. That is, the controller 30 can raise the boom 4 in response to the boom raising operation by the operator or independently of the boom raising operation by the operator.

[0105] Further, the controller 30 can supply the hydraulic fluid discharged from the pilot pump 15

to the right pilot port of the control valve **174R** via the proportional valve **31BR** in response to the boom lowering operation by the operator. Further, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the right pilot port of the control valve **174R** via the proportional valve **31BR**, independent of the boom lowering operation by the operator. That is, the controller **30** can lower the boom **4** in response to the boom lowering operation by the operator or independently of the boom lowering operation by the operator.

[0106] As illustrated in FIG. **4C**, the right operation lever **26R** is also used to operate the bucket **6**. Specifically, the right operation lever **26R** uses the hydraulic fluid discharged from the pilot pump **15** to apply a pilot pressure corresponding to the operation in the right-left direction to the pilot ports of the control valve **173R**.

[0107] Specifically, when the right operation lever **26R** is operated in the bucket closing direction (left direction), the right operation lever **26R** applies a pilot pressure corresponding to the operation amount to the left pilot port of the control valve **173R**. When the right operation lever **26R** is operated in the bucket opening direction (right direction), the right operation lever **26R** applies a pilot pressure corresponding to the operation amount to the right pilot port of the control valve **173R**.

[0108] The operation sensor **29RB** detects the operation content of the right operation lever **26R** in the left-right direction by the operator, and outputs the detected value to the controller **30**.

[0109] The proportional valve **31CL** operates in response to a control command (electric current command) output from the controller **30**. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the left pilot port of the control valve **173R** via the proportional valve **31CL**. The proportional valve **31CR** operates in response to a control command (electric current command) output from the controller **30**.

[0110] The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the right pilot port of the control valve **173R** via the proportional valve **31CR**. The proportional valve **31CL** can adjust the pilot pressure so that the control valve **173R** can be stopped at any position. Similarly, the proportional valve **31CR** can adjust the pilot pressure so that the control valve **173R** can be stopped at any position.

[0111] With this configuration, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the left pilot port of the control valve **173R** via the proportional valve **31CL** in response to the bucket closing operation by the operator. The controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the left pilot port of the control valve **173R** via the proportional valve **31CL**, independent of the bucket closing operation by the operator. That is, the controller **30** can close the bucket **6** in response to the bucket closing operation by the operator or independently of the bucket closing operation by the operator.

[0112] Further, the controller **30** can supply the hydraulic fluid discharged by the pilot pump **15** to the right pilot port of the control valve **173R** via the proportional valve **31CR** in response to the bucket opening operation by the operator. The controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the right pilot port of the control valve **173R** via the proportional valve **31CR**, independent of the bucket opening operation by the operator. That is, the controller **30** can open the bucket **6** in response to the bucket opening operation by the operator or independently of the bucket opening operation by the operator.

[0113] As illustrated in FIG. **4D**, the left operation lever **26L** is also used to operate the slewing mechanism **2**. Specifically, the left operation lever **26L** uses the hydraulic fluid discharged from the pilot pump **15** to apply a pilot pressure corresponding to the operation in the left-right direction to the pilot port of the control valve **173L**.

[0114] Specifically, when the left operation lever **26L** is operated in the left slewing direction (left direction), the left operation lever **26L** applies a pilot pressure corresponding to the operation amount to the left pilot port of the control valve **173L**. When the left operation lever **26L** is operated in the right slewing direction (right direction), the left operation lever **26L** applies a pilot pressure

corresponding to the operation amount to the right pilot port of the control valve **173L**.

[0115] The operation sensor **29LB** detects the content of the operation in the left-right direction performed on the left operation lever **26L** by the operator, and outputs the detected value to the controller **30**.

[0116] The proportional valve **31DL** operates in response to a control command (electric current command) output from the controller **30**. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the left pilot port of the control valve **173L** via the proportional valve **31DL**. The proportional valve **31DR** operates in response to a control command (electric current command) output from the controller **30**.

[0117] The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the right pilot port of the control valve **173L** via the proportional valve **31DR**. The proportional valve **31DL** can adjust the pilot pressure so that the control valve **173L** can be stopped at any position. Similarly, the proportional valve **31DR** can adjust the pilot pressure so that the control valve **173L** can be stopped at any position.

[0118] With this configuration, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the left pilot port of the control valve **173L** via the proportional valve **31DL** in response to the left slewing operation by the operator. Further, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the left pilot port of the control valve **173L** via the proportional valve **31DL**, independent of the left slewing operation by the operator. That is, the controller **30** can cause the slewing mechanism **2** to slew to the left, in response to the left slewing operation by the operator or independently of the left slewing operation by the operator.

[0119] Further, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the right pilot port of the control valve **173L** via the proportional valve **31DR** in response to the right slewing operation by the operator. Further, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the right pilot port of the control valve **173L** via the proportional valve **31DR**, independent of the right slewing operation by the operator. That is, the controller **30** can turn the slewing mechanism **2** to the right in response to the right slewing operation by the operator or independently of the right slewing operation by the operator.

[0120] Next, a machine guidance function and a machine control function of the excavator **100** will be described with reference to FIG. 5. FIG. 5 is a block diagram illustrating an example of a configuration related to a machine guidance function and a machine control function of the excavator.

[0121] The controller **30** executes control of the excavator **100** related to a machine guidance function of guiding (leading) manual operation of the excavator **100** by the operator, for example.

[0122] The controller **30** transmits work information such as a distance between the target construction surface and the distal end portion of the attachment, that is, the work portion of the end attachment to the operator through the display device **D1**, the sound output device **D2**, or the like.

[0123] Specifically, the controller **30** acquires information from the boom angle sensor **S1**, the arm angle sensor **S2**, the bucket angle sensor **S3**, the body inclination sensor **S4**, the slewing state sensor **S5**, the space recognition device **81**, the positioning device **84**, the input device **83**, and the like.

[0124] The data related to the target construction surface is stored in the internal memory, an external storage device connected to the controller **30**, or the like, for example, based on a setting input through the input device **83** by the operator or by being downloaded from the outside (e.g., a predetermined management server).

[0125] The data related to the design surface is expressed by, for example, a reference coordinate system. The reference coordinate system is, for example, a world geodetic system. The world geodetic system is a three dimensional orthogonal XYZ coordinate system with the origin at the center of gravity of the earth, the X-axis in the direction of the intersection of the Greenwich

meridian and the equator, the Y-axis in the direction of **90** degrees east longitude, and the z-axis in the direction of the north pole. For example, the operator may define any given point in the construction site as a reference point and set the design surface based on a relative positional relationship with the reference point through the input device **83**.

[0126] The work portion of the bucket **6** is, for example, the claw tip of the bucket **6**, the back surface of the bucket **6**, or the like. Further, when a breaker is employed as the end attachment instead of the bucket **6**, for example, the tip of the breaker corresponds to the work portion. Thus, the controller **30** can notify the operator of the work information through the display device **D1**, the sound output device **D2**, and the like, and can guide the operator to operate the excavator **100** through the operation device **26**.

[0127] Further, the controller **30** executes control of the excavator **100** related to a machine control function of supporting manual operation of the excavator **100** by an operator or automatically or autonomously operating the excavator **100**, for example. Specifically, the controller **30** is configured to acquire a target trajectory that is a trajectory followed by a position serving as a control reference (hereinafter, simply referred to as a “control reference”) set in a work portion or the like of the attachment.

[0128] In the case where there is a work target (e.g., the ground or the earth and sand on a bed of a dump truck described later) with which the end attachment can come into contact during work such as excavation work or rolling compaction work, a work portion of the end attachment (e.g., a claw tip or a back surface of the bucket **6**) may be set as the control reference. In addition, in the case of an operation in which there is no work target with which the end attachment can come into contact, such as a boom raising and slewing operation, an earth and sand discharge operation, or a boom lowering and slewing operation, which will be described later, any given portion (e.g., the lower end portion or the claw tip of the bucket **6**) that can define the position of the end attachment in the operation may be set as the control reference.

[0129] For example, the controller **30** derives the target trajectory based on the data related to the target construction surface. The controller **30** may derive the target trajectory based on the information on the terrain around the excavator **100** recognized by the space recognition device **81**. The controller **30** may derive information on a previous trajectory of the work portion such as the claw tip of the bucket **6** from a previous output of a posture detection device temporarily stored in the internal volatile storage device, and derive the target trajectory based on the information. The controller **30** may derive the target trajectory based on the current position of a predetermined portion of the attachment and the data related to the target surface.

[0130] The posture detection device includes, for example, the boom angle sensor **S1**, the arm angle sensor **S2**, the bucket angle sensor **S3**, the blade angle sensor **96**, and the like.

[0131] For example, when the operator manually performs excavation work, leveling work, or the like of the ground, the controller **30** automatically operates at least one of the boom **4**, the arm **5**, and the bucket **6** so that the target surface and a tip position of the bucket **6**, specifically, a work portion such as the claw tip or the back surface of the bucket **6**, are aligned with each other.

[0132] Specifically, when the operator performs an operation in the front-rear direction on the left operation lever **26L** while operating (pressing) the switch **SW**, the controller **30** automatically operates at least one of the boom **4**, the arm **5**, and the bucket **6** in response to the operation so that the target surface and the tip position of the bucket **6** are aligned with each other. More specifically, as described above, the controller **30** controls the proportional valves **31** to automatically operate at least one of the boom **4**, the arm **5**, and the bucket **6**. Thus, the operator can cause the excavator **100** to perform excavation work, leveling work, or the like along the target surface only by operating the left operation lever **26L** in the front-rear direction.

[0133] Further, for example, when a predetermined condition (hereinafter, referred to as an “excavation start condition”) is satisfied after the boom lowering and slewing operation of the excavator **100**, the controller **30** may automatically perform the excavation operation in accordance

with the operation of the attachment by the operator and move the bucket **6** in accordance with a predetermined target trajectory. The excavation start condition is a condition indicating the start of the excavation operation after the boom lowering and slewing operation of the excavator **100**. For example, the excavation start condition may include a condition that “an operation related to the arm **5** has been performed (i.e., the left operation lever **26L** has been operated in the front-rear direction) in a state where the bucket **6** is above the target surface”.

[0134] As described above, when a predetermined condition, that is, a condition corresponding to “operation on an operation target that has not been operated is started through a predetermined operation part (e.g., the operation device **26**)” is satisfied, the controller **30** automatically causes the excavator **100** to perform, in accordance with the operation on the operation target, a predetermined operation to move a predetermined portion of the attachment in accordance with the target trajectory.

[0135] When an operation on the attachment is input, the controller **30** of the present embodiment detects whether the person is present around the excavator **100**, and when a person is not present, the controller **30** performs a blade lowering operation of lowering the blade **95** to cause the blade **95** to be in contact with the ground, and prohibits the excavator **100** from traveling, based on the image or the like acquired by the space recognition device **81**. The controller **30** may also automatically perform the excavation operation in accordance with the operation of the operator related to the attachment.

[0136] Therefore, in the present embodiment, the excavation start condition of the excavator **100** may include a “state where no person is detected around the excavator **100** and the blade **95** is in contact with the ground” in addition to “an operation related to the arm **5** being performed (i.e., the left operation lever **26L** is operated in the front-rear direction) in a state where the bucket **6** is located above the target surface”.

[0137] Further, for example, when a predetermined condition (hereinafter, referred to as a “boom raising and slewing start condition”) is satisfied, the controller **30** automatically performs a raising operation of the boom **4** and the like in accordance with the slewing operation by the operator, and moves the bucket **6** along a predetermined target trajectory.

[0138] The boom raising and slewing start condition is a condition indicating the start of work for moving earth and sand or the like stored in the bucket **6** toward a dump truck parked at a predetermined position. For example, the boom raising and slewing start condition may include a condition that “the operation direction of the left operation lever **26L** has been switched from the front-rear direction to the right-left direction in a state where the machine control function is enabled, that is, in a state where the switch **SW** is pressed”, as will be described later.

[0139] Further, for example, the boom raising and slewing start condition may include a condition that “the left operation lever **26L** is operated in the left direction or in the right direction in a state where a predetermined switch (hereinafter, referred to as a “boom raising and slewing start switch”) provided at the distal end portion of the left operation lever **26L**, which can be included in the input device **83**, is pressed”. Further, for example, the boom raising and slewing start condition may include a condition that “the amount of earth and sand excavated by the attachment has become equal to or larger than a predetermined amount”.

[0140] Further, for example, the boom raising and slewing start condition may include “completion of excavation by the attachment by a predetermined distance or more”. In this case, the controller **30** can identify the amount of earth and sand excavated by the attachment, the excavation distance, and the Like based on an image of the front of the upper slewing body **3** captured by a monocular camera or a stereo camera that can be included in the space recognition device **81**, for example.

[0141] That is, the boom raising and slewing start condition is a condition for determining whether one operation of the excavator **100** such as the excavation operation is completed, for example. In addition, when the boom raising and slewing start condition includes a plurality of conditions as described above, the boom raising and slewing start condition may be satisfied when any one of the

plurality of conditions included is satisfied, or the boom raising and slewing start condition may be satisfied when some of two or more of the plurality of conditions included, or all of the plurality of the conditions included are satisfied.

[0142] The same applies to an earth and sand discharge start condition, a boom lowering slewing start condition, and the like, which will be described later. Specifically, when the operator operates the left operation lever **26L** in the left: direction or in the right direction, the controller **30** automatically operates at least the boom **4** of the upper slewing body **3** and the attachment in response to the operation so that the target trajectory and a portion (e.g., the lower end portion of the bucket **6**) serving as a control reference of the bucket **6** are aligned with each other.

[0143] More specifically, the controller **30** controls the proportional valves **31** to automatically operate the upper slewing body **3**, the boom **4**, and the like as described above. Thus, the operator can cause the excavator **100** to perform the boom raising and slewing operation for moving the earth and sand and the like stored in the bucket **6** to the dump truck by simply operating the left operation lever **26L** in the left-right direction.

[0144] Further, for example, when a predetermined condition (hereinafter, referred to as an “earth and sand discharge start condition”) is satisfied, the controller **30** automatically performs the opening operation of the arm **5** and the like in accordance with the opening operation of the bucket **6**, and discharges the earth and sand and the like stored in the bucket **6** toward the dump truck. The earth and sand discharge start condition is a condition indicating the start of work for discharging the earth and sand and the like stored in the bucket **6** to the dump truck.

[0145] For example, the earth and sand discharge start condition may include a condition that “the state where the left operation lever **26L** is operated in the left-right direction is switched to the state where the right operation lever **26R** is operated in the left-right direction (specifically, the left direction corresponding to the opening operation of the bucket **6**) in a state where the machine control function is enabled, that is, in a state where the switch **SW** is pressed”, as will be described later.

[0146] Further, for example, the earth and sand discharge start condition may include a condition that “the right operation lever **26R** is operated in the left direction (the operation of closing the bucket **6**) or in the right direction (the operation of opening the bucket **6**) in a state where a predetermined switch (hereinafter, referred to as “an earth discharge start switch”) provided at the tip portion of the right operation lever **26R**, which may be included in the input device **83**, is pressed”.

[0147] Further, for example, the earth and sand discharge start condition may include a condition that “the bucket **6** has reached a predetermined point (e.g., an end point of the target trajectory) above the dump truck”. In this case, the “predetermined point (the end point of the target trajectory)” in the earth and sand discharge start condition may be changed every time the earth and sand discharge is performed. Specifically, when the operator operates the right operation lever **26R** in the right direction, the controller **30** causes the bucket **6** to open and causes the arm **5** to open in response to the operation so that the earth and sand and the like in the bucket **6** are discharged to a predetermined target position on the bed of the dump truck.

[0148] More specifically, the controller **30** controls the proportional valves **31** to automatically operate the arm **5**, the bucket **6**, and the like as described above. Thus, the operator can discharge the earth and sand and the like stored in the bucket **6** onto the bed of the dump truck only by operating the right operation lever **26R** in the left-right direction (specifically, in the right direction).

[0149] Further, for example, when a predetermined condition (hereinafter, referred to as a “boom lowering slewing start condition”) is satisfied, the controller **30** automatically performs the lowering operation of the boom **4** and the like in accordance with the slewing operation by the operator, and moves the bucket **6** in accordance with a predetermined target trajectory.

[0150] The boom lowering slewing start condition is a condition indicating the start of work for

sleewably moving the attachment to the original position for performing excavation work or the like after discharging the earth and sand or the like in the bucket **6** onto the bed of the dump truck. For example, the boom lowering slewing start condition may include a condition that “the right operation lever **26R** is switched from a state of being operated in the left-right direction (specifically, the right direction) to a state of the left operation lever **26L** being operated in the left-right direction” as will be described later.

[0151] Further, for example, the boom lowering slewing start condition may include a condition that “the left operation lever **26L** is operated in the left direction or in the right direction in a state where a predetermined switch (hereinafter, referred to as a “boom lowering slewing start switch”) which can be included in the input device **83** and is provided at the tip portion of the left operation lever **26L** is pressed”. Further, for example, the boom lowering slewing start condition may include a condition that “there is no earth and sand falling from the bucket **6** onto the bed of the dump truck”.

[0152] In this case, the controller **30** can identify the amount of earth and sand or the like in the bucket **6** based on, for example, an image of the front of the upper slewing body **3** captured by a monocular camera or a stereo camera which can be included in the space recognition device **81**. Specifically, when the operator operates the left operation lever **26L** in the left direction or in the right direction, the controller **30** automatically operates at least the boom **4** of the upper slewing body **3** and the attachment in response to the operation so that the target trajectory and the portion serving as the control reference of the bucket **6** are aligned with each other.

[0153] More specifically, the controller **30** controls the proportional valves **31** to automatically operate the upper slewing body **3**, the boom **4**, and the like as described above. Thus, the operator can cause the excavator **100** to perform the boom lowering and slewing operation of moving the attachment to the original position for the excavation work or the like after discharging the earth and sand or the like stored in the bucket **6** to the bed of the dump truck by only operating the left operation lever **26L** in the lateral direction.

[0154] Further, for example, when a predetermined condition (hereinafter, referred to as a “bucket leveling operation start condition”) is satisfied before the boom lowering and slewing operation of the excavator **100**, the controller **30** may automatically perform an operation (hereinafter, referred to as a “bucket leveling operation”) for leveling earth and sand or the like loaded on the bed of the dump truck in response to the operation of the attachment by the operator and move the bucket **6** in accordance with a predetermined target trajectory.

[0155] The bucket leveling operation start condition is a condition indicating the start of the bucket leveling operation after the earth and sand or the like in the bucket **6** are discharged onto the bed of the dump truck. For example, the bucket leveling operation start condition may include a condition that “there are no earth and sand falling from the bucket **6** onto the bed of the dump truck”.

[0156] Further, for example, the bucket leveling operation start condition may include a condition that an “operation related to the arm **5** is performed (i.e., the left operation lever **26L** is operated in the front-rear direction) in a state where the bucket **6** is present above the bed of the dump truck”. In this case, the controller **30** may generate the target trajectory based on the shape of the bed of the dump truck, which is defined in advance and stored in an internal or external communicating nonvolatile storage device.

[0157] Further, when an operation instructing normal traveling is input after the operation on the attachment is ended, the controller **30** detects whether a person is present around the excavator **100**, based on the image or the like acquired by the space recognition device **81**, and when the person is not present, a blade raising operation of raising the blade **95** is performed.

[0158] In other words, when an operation instructing normal traveling is input after the excavation operation is ended and no person is detected around the excavator **100**, the controller **30** raises the blade **95** to cause the excavator **100** to travel.

[0159] The controller **30** of the present embodiment may automatically perform an operation

(hereinafter, referred to as a “normal traveling operation”) of moving the lower traveling body to the target position along a predetermined target trajectory. The controller **30** may also use the condition that the blade **95** is separated from the ground and the condition that the bucket **6** has risen to a predetermined height from the ground as the conditions for starting normal traveling. Further, the controller **30** may automatically perform the operation of raising the blade **95** and the operation of raising the attachment.

[0160] Further, when an operation for instructing the leveling travel is input after the operation on the attachment is ended, the controller **30** detects whether a person is present around the excavator **100** based on the image or the like acquired by the space recognition device **81**, and performs the blade lowering operation of lowering the blade **95** when the person is not present.

[0161] In other words, when the operation for instructing the leveling travel is input and no person is detected around the excavator **100**, the controller **30** lowers the blade **95** to cause the excavator **100** to travel. Further, the controller **30** may automatically perform an operation (hereinafter, referred to as a “leveling travel operation”) of moving the lower traveling body along a target trajectory set in a predetermined region in order to level the ground in the predetermined region by the blade. The controller **30** may set the leveling travel start condition to be that the blade **95** is in contact with the ground and that the bucket **6** has risen to a predetermined height from the ground. Further, the operation of lowering the blade **95** and the operation of raising the attachment may be automatically performed. The controller **30** controls the blade **95** and the attachment based on at least one of the detection values of the space recognition device **81** and the posture detection device.

[0162] The following description will be given on the assumption that the machine control function is enabled when the left operation lever **26L** and the right operation lever **26R** are operated in a state where the switch **SW** is pressed.

[0163] Although not illustrated in FIG. 5, the excavator **100** may include a communication device for transmitting and receiving information to and from an external device via a network or the like, and information received by the communication device may be input to the controller **30**. The controller **30** may transmit the information acquired by the controller **30** to an external device via the communication device. The external device may be a support device for supporting work by the excavator **100**, a management device for managing a work portion of the excavator **100**, or the like.

[0164] Next, the operation of the excavator **100** according to the present embodiment will be described with reference to FIG. 6. FIG. 6 is a flowchart illustrating the operation of the excavator.

[0165] In the excavator **100** according to the present embodiment, the controller **30** receives an input of an attachment operation command instructing an operation of the attachment (step **S601**). In other words, the excavator **100** receives an operation of the attachment by the operator.

[0166] Subsequently, the controller **30** determines whether a person is detected around the excavator **100** (step **S602**). Specifically, for example, the controller **30** may determine whether a person is detected around the excavator **100** using image data around the excavator **100** acquired by the space recognition device **81**.

[0167] When a person is detected in step **S602**, the controller **30** proceeds to step **S609** described later.

[0168] When no person is detected in step **S602**, the controller **30** performs a blade lowering operation to lower the blade **95** (step **S603**). In particular, the controller **30** performs control to extend the blade cylinder **95A**.

[0169] Subsequently, the controller **30** determines whether both ends of the tip portion **95a** of the blade **95** are in contact with the ground (step **S604**). The controller **30** may determine whether both ends of the tip portion **95a** are in contact with the ground based on, for example, a signal output from the blade angle sensor **96**, or may determine whether both ends of the tip portion **95a** are in contact with the ground based on an image captured by the space recognition device **81**.

[0170] In step **S604**, when both ends of the tip portion **95a** of the blade **95** are not in contact with

the ground, the process proceeds to step S609 described below.

[0171] In step S604, when both ends of the tip portion 95a of the blade 95 are in contact with the ground, the controller 30 prohibits the excavator 100 from traveling while the attachment is operating (step S605). Specifically, the controller 30 may disable the input by the travel lever.

[0172] Subsequently, after the operation on the attachment is ended, the controller 30 receives a travel command for instructing the excavator to travel (step S606). Specifically, the controller 30 receives an input from the travel lever in a state where there is no input of an operation command for instructing an operation of the attachment (no operation on the operation lever).

[0173] Subsequently, the controller 30 determines whether a person is detected around the excavator 100 (step S607). When no person is detected in step S607, the controller 30 performs a blade raising operation to raise the blade 95 (step S608). Specifically, the controller 30 performs control to contract the blade cylinder 95A.

[0174] When a person is detected in step S607, the controller 30 stops the operation of the excavator 100 (step S609).

[0175] At this time, the controller 30 may output an alarm or the like by the display device D1 or the sound output device D2.

[0176] In the present embodiment, when an operation on the attachment is received, the blade 95 is lowered and brought into contact with the ground. Therefore, in the present embodiment, while the attachment is being operated, the tip portion 95a of the blade 95 is in contact with the ground, and it is possible to reduce shaking or the like of the body during work.

[0177] In the present embodiment, when the attachment is not operated and the operation for instructing the traveling is performed, the blade 95 is raised. Therefore, in the present embodiment, for example, when the excavator 100 performs excavation work, the operator does not need to perform an operation of lowering the blade 95 every time the operator performs an operation of the attachment. In the present embodiment, when the excavator 100 travels after finishing the excavation work, the operator does not need to perform the operation of raising the blade 95.

[0178] Therefore, according to the present embodiment, the operation by the operator can be simplified, and the operability can be improved.

[0179] In particular, in the present embodiment, the blade 95 automatically lowers during excavation by the excavator 100 and automatically rises during traveling of the excavator 100. Therefore, the present embodiment is effective in a case where the excavation and the traveling are repeated during work such as work of digging a trench, and is capable of improving the work efficiency of the excavator 100.

[0180] Further, in the present embodiment, the blade 95 is operated only when no person is detected around the excavator 100. Therefore, in the present embodiment, for example, when a person enters a range that cannot be visually observed from the operator seated in the cabin 10, the person entered in the range can be detected, and safety can be improved.

[0181] In the present embodiment, for example, when deep digging work is performed in which the bucket 6 is located below the lower traveling body 1, the lower traveling body 1 may be slewed so that the blade 95 is located on the rear side when the direction in which the cabin 10 faces is the front side.

[0182] In this case, for example, the controller 30 may calculate the position of the bucket 6 based on the input of an operation command of the attachment, and when the bucket 6 is located below the lower traveling body 1, the controller 30 may instruct the operator to slew the lower traveling body 1 so that the blade 95 is located on the rear side. Specifically, the controller 30 may cause the display device D1 to display a message for instructing the operator to slew the lower traveling body 1 so that the blade 95 is located on the rear side, or may cause the sound output device D2 to output a sound guidance.

[0183] In the present embodiment, as described above, when the deep digging work is performed, the blade 95 is located on the rear side, and thus it is possible to prevent the blade 95 and the boom

4 from coming into contact with each other.

[0184] In the present embodiment, when the blade **95** is raised in response to reception of the travel instruction, the blade **95** may be raised to a height at which the tip of the blade **95** is not in contact with the ground, and the ground leveling work may be performed. Whether to perform the ground leveling work in response to reception of the travel instruction may be set in advance by the operator.

[0185] Next, a case where the blade **95** is lowered in the excavator **100** according to the present embodiment will be described with reference to FIGS. 7A and 7B. FIGS. 7A and 7B include views illustrating a situation when the blade is lowered. FIG. 7A is a view illustrating a state where the tip portion **95a** of the blade **95** is in contact with the ground, and FIG. 7B is a view illustrating a state during the deep excavation work.

[0186] In the present embodiment, as illustrated in FIG. 7A, the blade **95** is lowered until both ends of the tip portion **95a** of the blade **95** are in contact with the ground **G1** which is the traveling surface by the excavator **100**. In the present embodiment, when it is detected that both ends of the tip portion **95a** of the blade **95** are in contact with the ground **G1**, the operation of the attachment is started.

[0187] Therefore, according to this embodiment, the ground contact area between the body of the excavator **100** and the ground **G1** is increased even when, for example, a groove **G2** is formed in the ground **G1**. Therefore, according to this embodiment, excavation work can be performed with the body in a stable condition, and work efficiency can be improved.

[0188] Further, in the excavator **100**, for example, when the deep digging work in which the bucket **6** is located below the lower traveling body **1** is performed, as indicated by an oval **97** of FIG. 7B, the blade **95** and the boom cylinder **7** may come into contact with each other.

[0189] In the excavator **100** according to the present embodiment, when performing the deep digging work, the lower traveling body **1** is slewed so that the blade **95** is in a rearward position, and thus it is possible to avoid the blade **95** and the boom cylinder **7** from coming into contact with each other.

[0190] In the present embodiment, for example, the distance between the boom cylinder **7** and the blade **95** may be calculated based on the image captured by the space recognition device **81F**, and when the distance is equal to or less than a certain value, a warning may be given to avoid the contact.

[0191] The distance between the boom cylinder **7** and the blade **95** may be calculated from, for example, the detection value of the blade angle sensor **96** and the detection value of the boom angle sensor **S1**. The distance between the boom cylinder **7** and the blade **95** may be calculated based on, for example, image data captured by an imaging device provided around the excavator **100**. The imaging device provided around the excavator **100** may be, for example, a fixed-point camera or the like installed at a work portion, or may be an imaging device provided in a flying object flying around the excavator **100**.

[0192] In the present embodiment, the controller **30** of the excavator **100** executes the processes illustrated in FIG. 6, but the present disclosure is not limited thereto.

[0193] The controller **30** of the present embodiment may be, for example, a control part provided in an external device outside the excavator **100**.

[0194] Specifically, the controller **30** may be implemented by a control part of a support device for remotely controlling the excavator **100** or a management device that manages the excavator **100**.

[0195] In this case, the support device or the management device transmits an operation command of the attachment or the travel command to the excavator **100** in response to an operation of the excavator **100** by the remote operator. In the excavator **100**, when receiving these commands, the controller **30** causes the excavator **100** to perform an operation according to the corresponding command.

[0196] Therefore, according to the present embodiment, even when the excavator **100** is remotely

operated, the excavator **100** can be caused to automatically raise and lower the blade **95**, and operability can be improved.

[0197] The scope of the present disclosure is not limited to the technical matters described in the above-described embodiment, and design changes and the like within the scope not departing from the gist of the present disclosure are also included in the scope of the present disclosure.

Claims

1. A control device for an excavator, the excavator including an upper slewing body, an attachment provided on the upper slewing body, a slewing mechanism, a lower traveling body, and a blade provided on the lower traveling body, the control device comprising: circuitry configured to receive an input of an operation command for instructing an operation of the attachment, determine whether a person is detected around the excavator, and cause the excavator to perform an operation of lowering the blade when the person is not detected.
 2. The control device according to claim 1, wherein the circuitry is configured to prohibit the excavator from traveling while performing the operation of lowering the blade.
 3. The control device according to claim 1, wherein the circuitry is configured to receive an input of a travel command for instructing the excavator to travel, and cause the excavator to perform an operation of raising the blade.
 4. The control device according to claim 3, wherein the circuitry is configured to determine whether both ends of a tip portion of the blade are in contact with a ground after the operation of lowering the blade is performed, and output an alarm when the both ends are not in contact with the ground.
 5. The control device for an excavator according to claim 4, wherein the circuitry is configured to output an alarm for avoiding contact between the blade and the attachment in accordance with a distance between the blade and the attachment when the operation indicated by the operation command is an operation by which an end attachment included in the attachment is located below the lower traveling body.
 6. The control device for an excavator according to claim 1, wherein the circuitry is configured to receive an input of a travel command for instructing the excavator to travel, and cause the attachment to perform an operation of raising the attachment.
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